Bonded anchor pryout test model

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1 Introduction

Regarding the pryout tests for bonded anchors we propose the model in Figure 1) for the process of virtual approval.

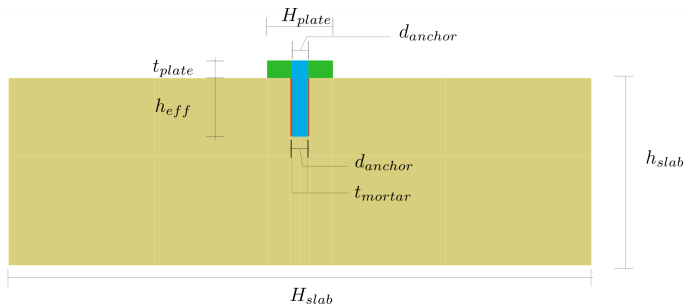

Figure 1: Geometry of the bonded anchor model

1.1 Geometry

The model consists of a cylindrical steel anchor with diameter d_{anchor} embedded till depth h_{eff} in an orthogonal parallelepiped concrete slab with dimensions (length: H_{slab}, width: H_{slab}, height: $2 * h_{eff}$). We cut a hole inside the concrete slub with diameter $d_{hole} = d_{anchor} + t_{mortar}$ and t_{mortar} is the mortar thickness. We do not model the drilling process or any of the residual stresses that may develop. The anchor is placed inside the hole so that their axes along the height match. The mortar region is then modeled as a hollow cylinder sleeve with height h_{eff} and constant thickness t_{mortar} between the anchor and the surrounding concrete.

The anchor is connected to a steel plate on top with dimensions (length: $H_{plate} = 4d_{anchor}$, width: $H_{plate} = 4d_{anchor}$, height: t_{plate}). The plate contains a hole in the middle with diameter matching the diameter of the anchor, through its thickness. The connection of the anchor to the plate is as follows: First, the anchor is extended above the concrete slab by a length equal to the thickness of the steel plate (i.e $h_{anchor} = h_{eff} + t_{plate}$). Second, the anchor extension is placed in the center of the steel plate hole. The two components (anchor, plate) then are merged together behaving as one body for subsequent discretizations of the model.

The assumed loading and the model are symmetric regarding the global z coordinate thus only half of the model needs to be simulated.
the component dimensions are summarized in the next table:

1.2 Finite element discretization

The model is constructed and discretized with the help of `Cubit` [1] see Figure **??** and then exported to `Abaqus` [2] format.

We use 20 node quadratic brick finite elements with reduced integration for the concrete (C3D20R), linear 8 noded brick finite elements for the steel plate and anchor and linear 8 noded cohesive elements

Component	Parameter	Description
Anchor	d_{anchor}	Anchor diameter
	h_{eff}	Effective embedment depth
	$h_{\text{anchor}} = h_{\text{eff}} + t_{\text{plate}}$	Total anchor height
Mortar	t_{mortar}	Mortar thickness
	$d_{\text{hole}} = d_{\text{anchor}} + t_{\text{mortar}}$	Drilled hole diameter
Concrete slab	$H_{\text{slab}} = 10h_{eff}$	Slab length and width
	$h_{\text{slab}} = 4h_{\text{eff}}$	Slab height
Steel plate	$H_{\text{plate}} = 4d_{\text{anchor}}$	Plate length and width
	$t_{\text{plate}} = d_{\text{anchor}}$	Plate thickness

Table 1: Geometrical parameters of the bonded anchor model

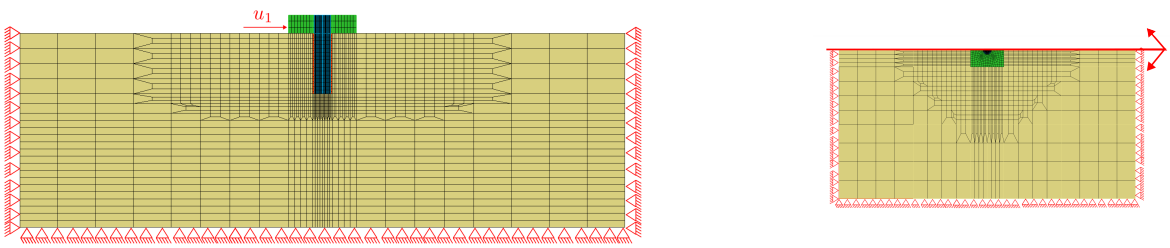

Figure 2: Mesh and boundary conditions of the bonded anchor model. Left: profile along the symmetry plane. Right: top view with symmetry condition applied.

(COH3D8) for the mortar.

The connection between the steel plate and the concrete is modeled by a hard contact interface law, i.e interpenetration is not allowed and the friction between the two bodies is not taken into account. The lateral connection between 1) anchor and mortar and 2) mortar and concrete is performed using tie constraints. The base of the anchor at the embedment height is free.

1.2.1 Material properties

The concrete is given with an elastic toy value. The advanced Gradient enhanced Concrete Damage Plasticity model (GCDP) is used in the full simulations with the corresponding user defined finite element.

Structural steel is modeled with its known elastic properties.

Mortar is modeled with the use of traction-separation law available in Abaqus including damage.
For all materials we use typical density properties. In order to take advantage of the implicit dynamic procedure later on, we increase the density uniformly by one order of magnitude (x10).

1.3 Boundary conditions

A uniform horizontal displacement boundary condition is applied on the left part of the steel plate at $t_{plate}/3$ from the concrete surface. Other than that the plate is left free to rotate due to the eccentric displacement field applied.

The concrete slab is laterally constrained in all directions at its boundaries.

1.4 Solution procedure

We perform an implicit dynamic analysis using the Backward Euler solver of Abaqus, in one step with adaptive time incrementation.

References

[1] Coreform LLC. Coreform cubit, 2025. Computer software.

[2] Dassault Systèmes SIMULIA. *Abaqus 2024 Documentation*. Dassault Systèmes, 2024. Finite element analysis software.